

CBSE Test Paper 03
Chapter 11 Thermal Properties of Matter

1. The triple points of carbon dioxide is 216.55 K. Express this temperatures on the Celsius and Fahrenheit scales **1**
 - a. $-66.60^{\circ}\text{C} = -99.88^{\circ}\text{F}$
 - b. $-86.60^{\circ}\text{C} = -150.88^{\circ}\text{F}$
 - c. $-56.60^{\circ}\text{C} = -69.88^{\circ}\text{F}$
 - d. $-76.60^{\circ}\text{C} = -120.88^{\circ}\text{F}$
 2. A brass wire 1.8 m long at 27°C is held taut with little tension between two rigid supports. If the wire is cooled to a temperature of -39°C , what is the tension developed in the wire, if its diameter is 2.0 mm? Co-efficient of linear expansion of brass = $2.0 \times 10^{-5} \text{ K}^{-1}$; Young's modulus of brass = $0.91 \times 10^{11} \text{ Pa}$. **1**
 - a. $3.2 \times 10^2 \text{ N}$
 - b. $3.4 \times 10^2 \text{ N}$
 - c. $3.8 \times 10^2 \text{ N}$
 - d. $3.6 \times 10^2 \text{ N}$
 3. A 10 kW drilling machine is used to drill a bore in a small aluminum block of mass 8.0 kg. How much is the rise in temperature of the block in 2.5 minutes, assuming 50 percent of power is used up in heating the machine itself or lost to the surroundings. Specific heat of aluminum = $0.91 \text{ J g}^{-1}\text{K}^{-1}$. **1**
 - a. 103°C
 - b. 93°C
 - c. 113°C
 - d. 123°C
 4. A brass rod of length 50 cm and diameter 3.0 mm is joined to a steel rod of the same length and diameter. What is the change in length of the combined rod at 250°C , if the original lengths are at 40.0°C ? The ends of the rod are free to expand (Co-efficient of linear expansion of brass = $2.0 \times 10^{-5} \text{ K}^{-1}$, steel = $1.2 \times 10^{-5} \text{ K}^{-1}$). **1**
 - a. 0.38 cm.
 - b. 0.42 cm.
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- c. 0.34 cm.
- d. 0.36 cm.
5. On a day when the temperature reaches 50°F , what is the temperature in degrees Celsius and in kelvins? **1**
- a. 12°C , 303 K
- b. 8°C , 273 K
- c. 10°C , 283 K
- d. 14°C , 323 K
6. Why should a thermometer bulb have a small heat capacity? **1**
7. Explain why do detergents have small angle of contact? **1**
8. If all the objects radiate electromagnetic energy, why do not the objects around us in everyday life become colder and colder? **1**
9. If a drop of water falls on a very hot iron, it does not evederate for a long time. Give reason. **2**
10. The difference between length of a certain brass rod and that of a steel rod is claimed to be constant at all temperatures. Is this possible? **2**
11. When we step barefoot into an office with a marble floor, we feel cold. Why? **2**
12. Show that the coefficient of volume expansion for a solid substance is three times its coefficient of linear expansion. **3**
13. Calculate the stress developed inside a tooth cavity filled with copper when hot tea at temperature of 57°C is drunk. You can take body (tooth) temperature to be 37°C and $\alpha = 1.7 \times 10^{-5}/\text{K}$. Bulk modulus for copper = $140 \times 10^9 \text{N/m}^2$. **3**
14. A copper block of mass 2.5 kg is heated in a furnace to a temperature of 500°C and then placed on a large ice block. What is the maximum amount of ice that can melt? (Specific heat of copper = $0.39 \text{ J g}^{-1} \text{ K}^{-1}$; heat of fusion of water = 335 Jg^{-1}). **3**
15. A brass wire of length 1.8 m at 27°C is held tight with little tension between two rigid supports. If the wire is cooled to a temperature of -39°C . What is the tension developed in the wire, if its diameter is 2.0 mm? Co-efficient of linear expansion of brass = $2.0 \times 10^{-5} \text{K}^{-1}$; Young's modulus of brass = $0.91 \times 10^{11} \text{Pa}$. **5**
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